

Muhammad Umar Khan

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SUMMARY

M.S. Computer Science student specializing in full-stack and AI-driven systems. Experienced in building scalable backend architectures, REST APIs, and LLM-integrated applications for document processing and automation. Proficient in Java, Python, JavaScript, and C#, with a focus on production-ready software engineering.

EDUCATION

Wayne State University | Detroit, MI, USA

Jan. 2025 - Aug. 2026

M.S. in Computer Science | GPA: 3.92/4

University of Windsor | Windsor, ON, Canada

Sep. 2020 – Apr. 2024

B.S. in Computer Science with Software Engineering Specialization

TECHNICAL SKILLS

Languages: Java, Python, C#, JavaScript/TypeScript, SQL

Frameworks: Spring Boot, Spring Security, ASP.NET, MVC, .NET 10, FastAPI, Flask, Node.js, React, Next.js, Razor

Databases: PostgreSQL, MongoDB, SQLite

Tools & DevOps: Git, Docker, AWS (Elastic Beanstalk, RDS, S3, CloudFront), REST APIs, JWT Authentication, Junit, Redis, Celery, Railway, Vercel

Frontend: HTML, CSS, Tailwind CSS, Vite

EXPERIENCE

University of Windsor | Windsor, ON, Canada

Apr. 2023 - Nov. 2024

Research Assistant

- Developed a Python-based image segmentation pipeline using clustering to generate superpixels, reducing preprocessing time by ~20%.
- Evaluated Graph-CNN models for pixel classification, selecting optimal architecture based on accuracy and runtime.
- Designed and integrated SQL-based data workflows with Snowflake, and containerized applications using Docker.

University of Windsor | Windsor, ON, Canada

Sep. 2022 – Apr. 2024

Teaching Assistant

- Conducted weekly office hours, assisted lab sessions, and graded assignments/exams for 100+ students in Data Structures, Algorithms, Database Systems, and Object-Oriented Programming (Java).

PROJECTS

AutoOps AI | ASP.NET Core, C#, React, TypeScript, FastAPI, PostgreSQL, Redis, Celery, Docker, Railway, Vercel

- Built a full-stack AI document processing platform with an ASP.NET Core REST API, JWT authentication with refresh token flow, and Entity Framework Core for database access across a PostgreSQL backend.
- Designed a Redis-backed Celery worker pipeline processing documents through parse → chunk → analyze stages with real-time SSE progress updates streamed to the React frontend.
- Containerized the multi-service architecture using Docker Compose and deployed to Railway (backend/worker) and Vercel (frontend) with environment-driven CORS, JWT, and Redis configuration.

Blog Platform | Java, Spring Boot, React, TypeScript, PostgreSQL, AWS

- Deployed a full-stack blog platform on AWS using Elastic Beanstalk (backend), RDS PostgreSQL (database), S3 and CloudFront (frontend CDN), with all API traffic proxied through CloudFront for unified HTTPS delivery.
- Implemented JWT-based authentication with role-based access control (RBAC), restricting category/tag management to admin users and post editing to authors.
- Built a rich text editing experience using TipTap with heading levels, lists, and inline formatting, with DOMPurify sanitization on the read side.

AdaptIQ | React, FastAPI, Python, JavaScript, Google Gemini API, Railway

- Built a full-stack AI resume tailoring platform using a single-pass LLM pipeline for keyword extraction and ATS optimization, reducing API calls by 50%.
- Implemented JWT-based password authentication with a token middleware layer protecting all API routes, and deployed the full-stack app to Railway serving the React build as static files from FastAPI.
- Designed a 5-metric ATS scoring engine (parsing ability, keyword injection, coverage, quantification rate, job match) that programmatically scans tailored resume text and surfaces actionable improvement insights per generation.